

Education

New York University, New York, NY

Aug 2024 – May 2026

Master of Science in Computer Engineering (GPA: 3.9/4.0)

Coursework: Machine Learning, Deep Learning, Big Data, System Design, Computer Architecture, Data Structures

Experience

- **CampusMesh** Jan 2026 – Present
AI Software Engineer
– Built **LLM-powered retrieval pipelines (RAG)** for CampusIQ (AI Assistant), applying **embedding-based semantic search** and structured data indexing to improve contextual accuracy by **30%** for clinician-adjacent academic workflows.
– Developed scalable **TypeScript/Node.js microservices** powering real-time workflows across messaging, calls, notifications, and profile management, backed by **MySQL (Sequelize)** and **DynamoDB**.
– Deployed **serverless architecture** on AWS with **Lambda, API Gateway, S3, and CloudFront** across **4 environments**; enforced secure access with **KMS-managed keys** and **least-privilege IAM**.
- **Amazon** May 2025 – Aug 2025
Software Development Engineer Intern
– Deployed a **Retrieval-Augmented Generation (RAG)** pipeline with an **agentic workflow** that retrieves operational logs to reason over incident details and generate context-aware remediation steps for production systems.
– Built a high-throughput incident triage system on **AWS ECS Fargate, SQS, SNS**, reducing response latency by **50%**; provisioned resilient **Infrastructure-as-Code** with **AWS CDK**, ensuring **99.9%** availability under peak load.
– Implemented a **Model Context Protocol (MCP)** server for tool-augmented retrieval, improving query efficiency and compute utilization; conducted code reviews on backend changes with **Mockito** test coverage.
- **New York University (Novel AI Technologies, Inc.)** Aug 2024 – Present
AI Software Engineer
– Architected an **NSF-funded (\$100K)** AI-powered **health platform** for **iOS** and **Android** ingesting real-time wearable telemetry; deployed cloud-hosted models for health metric analysis and patient-facing insights.
– Built **LLM-powered workflows (GPT-4 + LangChain)** for document analysis and a full-stack product (**React, TypeScript, Node.js, REST APIs**) with **RBAC** and row-level security for regulated healthcare-adjacent clients.
– Engineered **Python ETL pipelines** transforming **NoSQL** into **PostgreSQL**; processed **250K+** records and powered **AWS QuickSight** dashboards for operational monitoring.
- **PricewaterhouseCoopers (PwC)** Aug 2021 – Aug 2024
Senior Software Engineer
– Designed a distributed backend on **Microsoft Azure** for a **health platform**, supporting **ML inference pipelines** and integrations with **Microsoft Dynamics**.
– Built an **NLP-powered OCR pipeline** using **LayoutLM** and Azure Form Recognizer to automate structured data extraction, reducing processing time by **75%**.
– Led code reviews and mentored engineers on **software design and test-driven development**, reducing production bugs by **25%**.

Projects

- **InviLite (LLM-Powered Reliability Platform)** | *Python, AWS Bedrock, FAISS, Kinesis, Lambda, CloudWatch*
– Built a **RAG system (SVM + FAISS + Bedrock LLM)** over operational telemetry, cutting incident resolution from **1–2 hours to minutes** and lifting classification accuracy from **56% to 92%**.
– Architected cloud-native ingestion pipelines (**Kinesis, Lambda, S3, Glue, DynamoDB**) with monitoring hooks for production triage workflows.
- **Hospital Management System (Healthcare Full-Stack)** | *React, TypeScript, Node.js, MongoDB, AWS SageMaker*
– Built a **full-stack healthcare platform** managing appointments, patient records, and lab reports with **role-based access control** and API-driven workflows.
– Fine-tuned a **Hugging Face transformer-based model** reaching **88% accuracy** for health condition prediction, deployed on **AWS SageMaker** for real-time support.

Skills

- **Programming Languages:** TypeScript, Python, JavaScript (ES6), Java, C++, SQL
- **AI/ML:** PyTorch, LLM Fine-tuning, RAG (LangChain, FAISS), Transformers, NLP, Hugging Face, Bedrock, GPT-4, Embeddings, Deep Learning, Pandas, NumPy
- **Web & Backend:** React, Node.js, Express.js, REST API, GraphQL API, Microservices
- **Databases:** PostgreSQL, MySQL, MongoDB, DynamoDB
- **Cloud & DevOps:** AWS (Lambda, SageMaker, Bedrock, ECS, S3, CloudWatch, CDK, API Gateway, IAM), Azure (AI Services, Function Apps), Docker, Kubernetes, CI/CD
- **Tools & Platforms:** Git, Linux, Jupyter, Postman, Model Context Protocol (MCP)

Publications

Supervised ML System Based Segmentation and Classification of Stroke Using Deep Learning Techniques (*IEEE*)

An Aquila-optimized SVM Classifier for Diabetes Prediction (*IEEE*)

Certifications

AWS Cloud Practitioner

Stanford Machine Learning (Coursera)

Azure Data Fundamentals